DATA ITEM DESCRIPTION

Title: VERIFICATION TEST PLAN

Number: DI-NUSU-82341
AMSC Number: 10236
DTIC Applicable: No
Preparing Activity: DS

Approval Date: 20210331
Limitation: N/A
GIDEP Applicable: No
Project Number: NUSU-2021-003

Use/Relationship:

The Verification Test Plan applies to both parts of military standard MIL-STD-188-125, *High-Altitude Electromagnetic Pulse (HEMP) Protection for Ground-Based Facilities Performing Critical, Time-Urgent Missions*. This DID applies to fixed facilities in accordance with Part I (MIL-STD-188-125-1A) and to transportable systems in accordance with Part II (MIL-STD-188-125-2). This DID contains the format, content, and intended-use information for the data deliverable resulting from the verification-test tasks described in section 4.4.3 and Appendices B and C of each part of MIL-STD-188-125, as applicable. This DID also identifies the same for the threat-level illumination verification-test tasks described in Part II's Appendix D, as applicable.

The purpose of verification testing the HEMP-protection subsystem (HPS) is to demonstrate that the installed protection provides the required HEMP hardness. These tests are performed after the construction and acceptance testing are complete and after the equipment is installed and functioning to determine if the operational system suffers mission-aborting damage or upset due to simulated HEMP excitations. Verification is normally a Government-conducted test and is not part of a facility construction contract.

The Verification Test Plan describes the test-execution, data-recording, analysis, inspection, and reporting procedures to be used by the contractor to perform the verification testing. The plan provides the means for the Government to understand and duplicate methods used by the contractor to demonstrate the HPS' performance with respect to MIL-STD-188-125's verification-test requirements. The Department of Defense's HEMP-certification authority requires that a documented verification test plan be submitted as part of a HEMP-certification request package for a facility or system.

If a portion of the HPS' components has already been verification tested by a previous test program, then documentation of those previous tests, including plans, reports, and test data, is attached as an appendix.

Requirements:

1. Reference documents. The applicable issue of the documents cited herein, including their approval dates and any applicable amendments, notices, and revisions, shall be as specified in the contract. The Verification Test Plan's classification shall be in accordance with any relevant facility- or system-specific classification guides and also the Defense Threat Reduction Agency (DTRA) Security Classification Guide for DoD Electromagnetic Pulse (EMP) Programs and Activities. (DTRA's security classification guide is available by mail request to ATTN: RD-

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NTSA, Dr. William Linzey, PhD, Defense Threat Reduction Agency, 8725 John J. Kingman Road, MSC 6201, Fort Belvoir, VA 22060-6201.)

2. Format. Contractor format is acceptable.

3. Content. The Verification Test Plan shall document the test-execution, data-recording, analysis, inspection, and reporting procedures to be used to comply with MIL-STD-188-125's section 5.14 and Appendices B and C of Parts I and II, and also Appendix D of Part II, including:

a. Pulsed current injection (PCI) test procedures per Appendix B

b. Continuous wave immersion (CWI) test procedures per Appendix C

c. Threat-level illumination (TLI) test procedures per Appendix D

3.1 Administrative data. The Verification Test Plan shall contain an administrative section covering the following:

a. Contract number, CDRL number, and date

b. Authentication and certification by a qualified representative of the procuring activity for the test plan

3.2 Summary information. The Verification Test Plan shall summarize the following:

3.2.1 Introduction, to include:

a. General objectives of the verification test

b. A general description of the HPS' topology to be tested

c. Statement of any assumptions, or limitations associated with the planned verification testing

d. Summary of the verification test's pass and fail requirements per MIL-STD-188-125

3.2.2 Scope. A general description of the overall test-requirements matrix that is used to verify the HPS' compliance with MIL-STD-188-125's performance requirements, including the relative roles of tests, inspections, and analyses.

3.2.3 Methods of verification. A general description of the procedures that are used to verify the HPS' compliance with each verification test performance requirement listed in section 3.3 below.

3.2.4 Engineering factors. Any important engineering factors affecting the verification-test procedures, such as: physical-accessibility constraints of the facility's or system's topology, resource limitations, safety and security protocols, and precedent reports.

3.3 Detailed information. The overall Verification Test Plan shall describe in detail the test procedures and the associated test equipment. The Verification Test Plan shall be in three portions, one each for the PCI and CWI testing (as detailed in Appendices B and C of both Part I and Part II of MIL-STD-188-125) and one portion, where applicable, for the TLI testing (as detailed in Part

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II's Appendix D). These test-plan portions shall be detailed in three separate sections of the overall verification test plan. For the verification test of a transportable ground-based system, the test plan shall note whether TLI tests will be performed in lieu of CWI testing, or vice versa, and include an explanation for why this decision was made.

3.3.1 The overall test plan. The main portion of the test plan shall contain the following detailed information:

a. An identification of the facility or system to be tested

b. Identification of the proposed test-team organizations and personnel and of the Government organizations and witnesses that will likely be present

c. A description of the HPS' topology, to include: drawings showing the locations of all points of entry (POEs) and POE protective devices (PPDs) of each electromagnetic (EM) barrier, equipment layouts inside each low-risk EM barrier; the installation of mission critical systems (MCS) outside of each low-risk EM barrier and their associated special protective measures (SPM), cable interconnections external to each low-risk EM barrier, the presence of any long line protection modules (LLPMs) associated to transportable ground-based systems; and a description of each HEMP-protection measure

d. Descriptions of the data acquisition equipment, to include: the manufacturer, model, and serial numbers of each equipment item; detailed calibration procedures for the equipment; and calibration-traceability documentation

e. Descriptions of the test procedures, including: system, subsystem, and circuit configuration requirements; equipment operating states, test configuration diagrams; test point locations and descriptions; step-by-step procedures; and functional monitoring provisions

f. Descriptions of methods for measuring noise (instrumental and local or ambient)

g. A description of models, techniques, and tools used for preliminary analyses and predictions, and their specific application to the specific facility or system to be tested

h. A description of the methods that will be applied to quantify and evaluate any failures of the HPS' performance with respect to all pass and fail criteria

i. Identification of any deviations from the requirements of MIL-STD-188-125 that are anticipated, along with an explanation of why these deviations may be necessary

j. Descriptions of the data-management procedures that will be used, to include: data quality control procedures, data acceptability criteria, annotation and preservation of data records, and the methods to compare results to pass and fail criteria

k. Identification of potential safety risks and descriptions of procedures to mitigate these risks, including electromagnetic radiation hazards and electrical shock hazards

l. A description of the data-security and physical-security procedures that are applied

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m. A description of the planned test schedule, including a listing of the prioritization of measurements

n. A description of the methods that will be used to simulate the operational performance of MCS during the testing if the actual MCS operation during the testing would be impractical

o. A written confirmation that the required MIL-STD-461 testing of the first articles of all MCS installed outside the shield was, or will be, completed with results available prior to the verification test

3.3.2 The PCI portion of the test plan. This portion of the Verification Test Plan shall contain the following detailed information:

a. Detailed test procedures in accordance with MIL-STD-188-125 Appendix B's section B.4.7 and Tables B-II and B-III

b. Identification, with diagrams, of the planned locations of each HPS component to be subjected to PCI injection and an explanation for why these specific points were selected

c. Identification of the expected measurement-range sensitivities of the current-measurement probes and data acquisition system

d. A table listing the associated PCI drive levels per test point

e. Descriptions of the methods to be used to determine the onset of any MCS upset or damage, and the locations for which observations will be made

f. Identification of any special requirements of the facility operations which may impact HEMP testing

3.3.3 The CWI portion of the test plan. This portion of the Verification Test Plan, if applicable, shall contain the following detailed information:

a. Detailed test procedures in accordance with MIL-STD-188-125 Appendix C's section C.4.4

b. Descriptions of the CWI antenna's field map characteristics, and the methods to be used to monitor the incident CWI and relationship to these field map characteristics

c. Identification, with diagrams, of the proposed locations and orientations of the facility or system under test with respect to each position and polarization of the CWI transmit antenna, and the placement of other test elements with respect to the CWI antenna (e.g., MCS outside the low-risk EM barrier), which are required to achieve functionality

d. A table listing all planned test point locations

e. Identification of the following instrumentation locations, with an explanation for why these specific points were selected:

   i. Electric and magnetic field sensors

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ii. Reference sensor

iii. Representative MCS cable measurement test points both within, and external to, all electromagnetic barriers

iv. Other measurement points as required

f. Identification of the expected measurement-range sensitivities of the field probes, current probes, and data acquisition system

g. Identification of the equations to be used to verify the HPS' SE performance

h. A description of the algorithms to be used to extrapolate the CWI results to the HEMP-threat level environments

i. A description of any alternative test methods that must be used for verifying an electromagnetic barrier's performance if a CWI test will not be possible due to constraints posed by the site's or system's location or construction

3.3.4 The TLI portion of the test plan. This portion of the Verification Test Plan, if applicable, shall contain the following detailed information:

a. Detailed test procedures in accordance with MIL-STD-188-125-2 Appendix D's section D.4

b. Descriptions of the TLI antenna's field map characteristics, and the methods to be used to monitor the incident field and its relationship to these field map characteristics

c. Diagrams of the placement of subsystems and equipment under test in the TLI simulator's working volume and placement of other system elements that are required to achieve functionality with respect to the simulator working volume

d. Diagrams of the interconnecting cable layouts

e. A table listing all planned test point locations

f. Identification of the following instrumentation locations, with an explanation for why these specific points were selected

i. Electric and magnetic field sensors

ii. Reference sensor

iii. Representative MCS cable measurement test points both within, and external to, all electromagnetic barriers

iv. Other measurement points as required

g. Identification of the expected measurement-range sensitivities of the field probes, current probes, and data acquisition system

3.4 Other information sources. When other available information sources contain data required by the Verification Test Plan, these sources shall be referenced, rather than duplicated, within the test plan.

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